HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

Chapter 2: Op-Amp Basics

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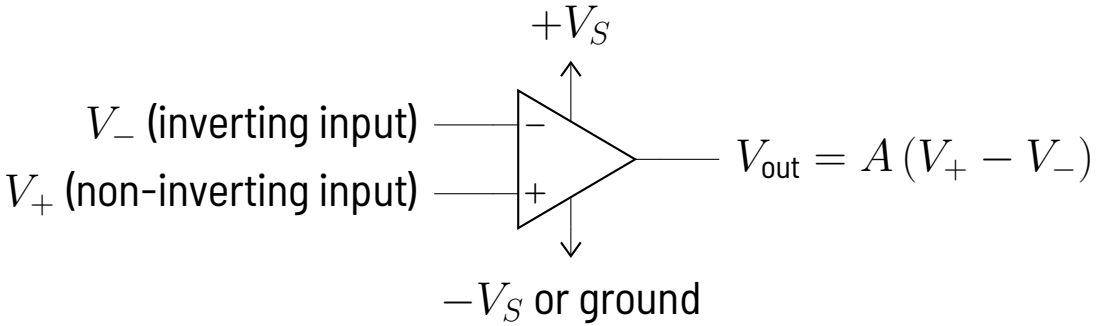
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2 Op-Amp Basics

2.1 The Op-Amp in a Nutshell

Symbol and model



- an amplifier for the **difference** of two voltages, with a huge gain $A = 10^5 \ldots 10^7$
- inputs draw (almost) no current; the output can deliver some 10 mA
- the output stays inside the supply range; supply pins are often not drawn

Negative feedback

- a part of V_{out} is fed back to the **inverting** input
- V_{out} rises $\Rightarrow V_-$ rises $\Rightarrow V_+ - V_-$ shrinks $\Rightarrow V_{\text{out}}$ is pulled back
- the circuit settles where $V_+ - V_- = V_{\text{out}}/A \approx 0$ ($50\,\mu\text{V}$ for $V_{\text{out}} = 5\,\text{V}$, $A = 10^5$)

Rule 2.1 ***Golden rules*** *for an op-amp with negative feedback:*

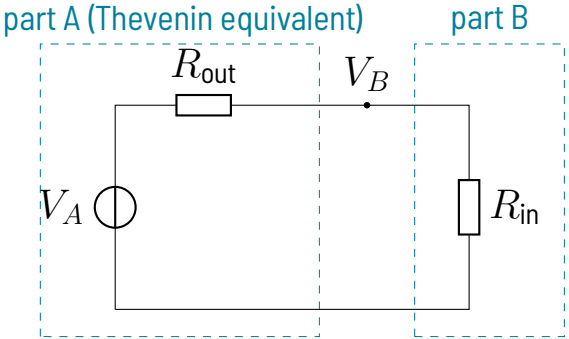
1. *The output does whatever is necessary to make the voltage difference between the inputs zero:* $V_+ = V_-$.
2. *The inputs draw no current:* $I_+ = I_- = 0$.

Real op-amps on a single supply

- most circuits in the lab run from a single supply (5 V or 12 V) and ground
- **input common-mode range**: the input voltages must stay inside it (often the supply minus 1.5 V)
- **output swing**: a rail-to-rail output reaches the rails within some 10 mV, older types (LM358) only within about 1.5 V of the positive rail
- **output current**: typically 10 mA … 40 mA, short-circuit protected
- examples: MCP6002, TLV9002 (CMOS, rail-to-rail, 5 V), LM358 (bipolar, up to 32 V)

2.2 The Unity-Gain Buffer

Every connection is a voltage divider

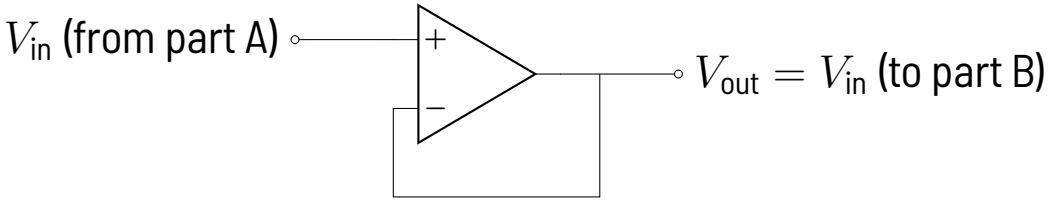


Loading

$$V_B = V_A \frac{R_{\text{in}}}{R_{\text{in}} + R_{\text{out}}} \qquad \text{error} \approx \frac{R_{\text{out}}}{R_{\text{in}}} \text{ for } R_{\text{in}} \gg R_{\text{out}}$$

- rule of thumb: $R_{\text{in}} \geq 100\, R_{\text{out}}$ for an error below $1\,\%$

The voltage follower



- rule 1: $V_- = V_+$, and $V_- = V_{\text{out}} \Rightarrow V_{\text{out}} = V_{\text{in}}$ (gain 1)
- rule 2: input current ≈ 0, input resistance GΩ ...TΩ: part A is **not loaded**
- output resistance with feedback $r_o/(1+A)$: $50\,\Omega/10^5 = 0.5$ mΩ: part B sees an **ideal source**

Unity-gain buffer

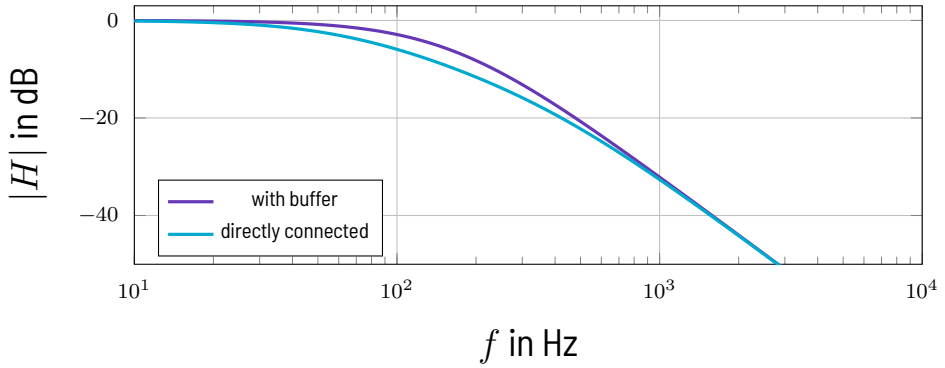
$V_{\text{out}} = V_{\text{in}} \qquad R_{\text{in}} \to \infty \qquad R_{\text{out}} \to 0$

Example: potentiometer as a set-point

- potentiometer $10\,\text{k}\Omega$ at $5\,\text{V}$, wiper in the middle: $V_A = 2.5\,\text{V}$, $R_{\text{out}} = 5\,\text{k}\Omega \parallel 5\,\text{k}\Omega = 2.5\,\text{k}\Omega$
- load $1\,\text{k}\Omega$ directly at the wiper: $V_B = 2.5\,\text{V} \cdot \frac{1}{1+2.5} = 0.714\,\text{V}$
- load behind a follower: $2.50\,\text{V}$; the op-amp delivers $2.5\,\text{mA}$
- the wiper itself still has $2.50\,\text{V}$: the follower does not load it

▶ Falstad: `examples/opampbasics/buffer_pot.txt`

Example: two RC low-passes in cascade



- each stage $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$: $f_c = 159\,\text{Hz}$
- with buffer: $H = H_1 \cdot H_2 = \frac{1}{(1+j\omega RC)^2}$, at f_c: $|H| = 1/2$ (-6 dB)
- directly: $H = \frac{1}{1+3j\omega RC-(\omega RC)^2}$, at f_c: $|H| = 1/3$ (-9.5 dB)

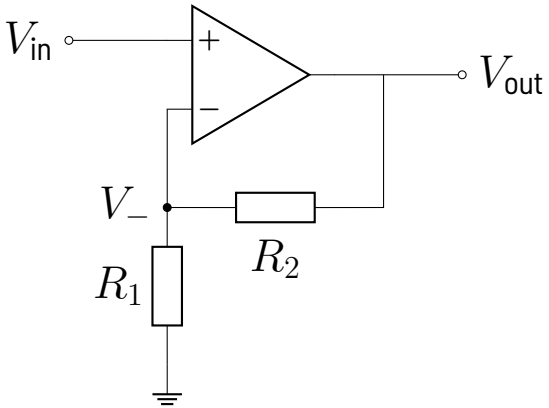
▶ Falstad: `examples/opampbasics/buffer_rc_cascade.txt`

Where buffers are used – and their limits

- in this lecture: behind a voltage reference (chapter 4), at bridge taps (chapter 5), between filter stages (chapter 7), behind the hold capacitor (chapter 8)
- limits
  - output current (some 10 mA) and output swing
  - input common-mode range: e.g. LM358 inputs only up to $V_S - 1.5$ V
  - capacitive loads (long cables): the follower may oscillate; add $50\,\Omega$ … $100\,\Omega$ in series with the output
  - offset voltage: $V_{\text{out}} = V_{\text{in}} + V_{\text{OS}}$ (µV …mV)

2.3 The Non-Inverting Amplifier

Non-inverting amplifier



- R_1, R_2 divide the output: $V_- = V_{\text{out}} \frac{R_1}{R_1+R_2}$; rule 1: $V_- = V_{\text{in}}$
- $R_2 = 0$ (or $R_1 \to \infty$): the follower with $G = 1$

Non-inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1}$$ input resistance $\approx \infty$, output resistance ≈ 0

Exercise 2.1: Buffering a potentiometer

A $10\,\mathrm{k}\Omega$ potentiometer between $5\,\mathrm{V}$ and ground sets a voltage. The wiper is in the middle position. The voltage is used by a following stage with an input resistance of $R_L = 1\,\mathrm{k}\Omega$.

a) Draw the Thevenin equivalent of the wiper. In which wiper position is R_{out} largest?

b) Calculate the voltage at R_L without a buffer and the error in percent.

c) How large must R_L be for an error below $1\,\%$ without a buffer?

d) Now a voltage follower (op-amp at $5\,\mathrm{V}$, rail-to-rail) is placed between wiper and R_L. Calculate the voltage at R_L, the current from the wiper and the output current of the op-amp.

e) The wiper is turned to the upper end ($5\,\mathrm{V}$). What does an LM358 at $5\,\mathrm{V}$ supply deliver, and why?

f) Check (b) and (d) with Falstad.

Solution 2.1

a) $V_A = 2.5\,\text{V}$, $R_{\text{out}} = 5\,\text{k}\Omega \parallel 5\,\text{k}\Omega = 2.5\,\text{k}\Omega$. For a wiper position x (0 …1): $R_{\text{out}} = x(1-x)\,10\,\text{k}\Omega$, largest in the middle ($R/4$).

b) $V_B = 2.5\,\text{V} \cdot 1\,\text{k}\Omega/(1\,\text{k}\Omega + 2.5\,\text{k}\Omega) = 0.714\,\text{V}$, error $-71\,\%$.

c) $R_L \geq 100 \cdot 2.5\,\text{k}\Omega = 250\,\text{k}\Omega$.

d) $V_B = 2.50\,\text{V}$; wiper current ≈ 0 (picoamperes for a CMOS op-amp); op-amp output current $2.5\,\text{V}/1\,\text{k}\Omega = 2.5\,\text{mA}$, which comes from the $5\,\text{V}$ supply.

e) The input common-mode range of the LM358 ends at about $V_S - 1.5\,\text{V} = 3.5\,\text{V}$, and its output does not reach the positive rail either: the output stays at about $3.5\,\text{V}$. For signals up to the supply a rail-to-rail op-amp is needed.

f) ▶ Falstad: `examples/opampbasics/buffer_pot.txt`

`V_direct` $= 0.714\,\text{V}$, `V_wiper` $=$ `V_buffered` $= 2.50\,\text{V}$.